Three Cubic Series for Generalized Central Trinomial Coefficients

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Abstract

We prove three evaluations of cubic generating series of generalized central trinomial coefficients. An arcsine integral represents each series as a real period of a double cover branched over six lines. Quadratic transformations identify the actual integration regions and their first parameter derivatives. Two evaluations then follow from classical elliptic identities. For the third required period, a product formula is established by an Euler–Poisson–Darboux equation with analytic Cauchy data, and a local elliptic correspondence gives the exact derivative evaluation. The algebraic transformations are accompanied by exact symbolic certificates.

Keywords: generalized central trinomial coefficients; Ramanujan-type series; elliptic integrals; period transformations.

1 Statement and conventions

For $n \in \mathbb{Z}_{\geq 0}$, define

$$T_n(b,c) = [X^n](X^2+bX+c)^n = \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j}\binom{2j}{j} b^{n-2j}c^j.$$

All square roots of positive real numbers are positive. In the complex arguments used in Appendix A, branches and continuation paths are specified explicitly. The notation $K(m)$ always uses the elliptic *parameter* m, rather than the modulus $\sqrt{m}$.

Theorem 1.1. *The following three series converge absolutely and satisfy*

$$\sum_{n=0}^{\infty} \frac{66n+17}{55296^n} T_n(10,121)^3 = \frac{540\sqrt{2}}{11\pi}, \tag{1.1}$$

$$\sum_{n=0}^{\infty} \frac{126n+31}{(-80)^{3n}} T_n(22,441)^3 = \frac{880\sqrt{5}}{21\pi}, \tag{1.2}$$

$$\sum_{n=0}^{\infty} \frac{3990n+1147}{(-288)^{3n}} T_n(62,9025)^3 = \frac{432(195\sqrt{14}+94\sqrt{2})}{95\pi}. \tag{1.3}$$

The proof distinguishes derivatives tangent to an elliptic period family from derivatives that move its integration boundaries. This distinction is needed even when a central half-rectangle has, at one parameter value, half the value of the complete rectangle.

2 Convergence and an integral representation

Lemma 2.1. *If $c > 0$, then*

$$T_n(b,c) = \frac{1}{\pi}\int_0^\pi (b + 2\sqrt{c}\cos\theta)^n \,\mathrm{d}\theta. \tag{2.1}$$

Consequently $|T_n(b,c)| \le (|b| + 2\sqrt{c})^n$.

Proof. Expand the integrand. Odd powers of $\cos\theta$ integrate to zero, whereas

$$\frac{1}{\pi}\int_0^\pi \cos^{2j}\theta \,\mathrm{d}\theta = 4^{-j}\binom{2j}{j}.$$

The displayed coefficient formula follows term by term. □

For the three series in theorem 1.1, the resulting geometric bounds are 16/27, 64/125, and 343/512, respectively. Each is less than one. These bounds, with a slightly larger ratio still less than one, hold in neighborhoods of the three parameter points. The series, their linearly weighted series, and the first derivatives used below therefore converge uniformly there.

Put

$$\beta = b + 2\sqrt{c}, \qquad r = \frac{b - 2\sqrt{c}}{\beta}, \qquad T = \beta^3 t, \qquad \mathcal{F}_r(T) = \sum_{n\ge 0} T_n(b,c)^3 t^n, \qquad \vartheta = T\frac{\mathrm{d}}{\mathrm{d}T}.$$

Here r is fixed when ϑ acts. The three pairs (r,T) are

$$(-3/8, 16/27), \qquad (-5/16, -64/125), \qquad (-32/63, -343/512). \tag{2.2}$$

Using theorem 2.1 three times, summing a geometric series, and integrating the third angular variable gives

$$\mathcal{F}_r(T) = \frac{1}{\pi^2}\int_r^1\int_r^1 \frac{\mathrm{d}x\,\mathrm{d}y}{\sqrt{(x-r)(1-x)(y-r)(1-y)(1-rTxy)(1-Txy)}}. \tag{2.3}$$

The one-variable integration uses $\pi^{-1}\int_0^\pi (A + B\cos\theta)^{-1}\,\mathrm{d}\theta = (A^2 - B^2)^{-1/2}$ with $A > |B|$; this inequality follows from $|T| < 1$ and $-1 < r < 0$. On setting $s = xy$, the absolute differential form becomes

$$\frac{|\,\mathrm{d}x\,\mathrm{d}s|}{\pi^2\sqrt{f(x,s)}}, \qquad f = (x-r)(x-1)(s-rx)(s-x)(1-rTs)(1-Ts). \tag{2.4}$$

The region is still specified by $r < x < 1$ and $r < s/x < 1$. The excluded line $x = 0$ has measure zero.

3 The transformed real integration regions

Let $a_i^\pm = \delta_i \pm 1$, and work on the real domain

$$q > 0, \qquad |\delta_i| < 1, \qquad q\prod_{i=1}^3 (1 + |\delta_i|) < 1. \tag{3.1}$$

Set

$$D_\delta(x,y) = [(x-\delta_1)^2 - 1][(y-\delta_2)^2 - 1][(1 - q\delta_3 xy)^2 - q^2x^2y^2]. \tag{3.2}$$

Every region integral in this section uses $\pi^{-2}\,\mathrm{d}x\,\mathrm{d}y/\sqrt{D_\delta(x,y)}$, with its positive real denominator. Define H_- on $a_1^- < x < a_1^+$, $a_2^- < y < 0$, and H_+ on $0 < x < a_1^+$, $a_2^- < y < a_2^+$. Define $J^+(q;\delta)$ on

$$x > a_1^+, \qquad y > a_2^+, \qquad z = \frac{1}{qxy} > a_3^+.$$

In multiplicative coordinates this integral is symmetric in the three centers. Changing the signs of the second and third variables gives

$$J^{+--}(q;\delta) = J^+(q;\delta_1,-\delta_2,-\delta_3).$$

Finally put

$$P_- = H_- + J^{+--}, \qquad P_+ = H_+ + J^{+--}, \qquad P_0 = H_+ + J^+. \tag{3.3}$$

Lemma 3.1. *These region integrals are real analytic on* (3.1).

Proof. For central regions use $x = \delta_1 + \cos\theta$ and $y = \delta_2 + \cos\phi$. The truncated endpoints are $\arccos(-\delta_i)$, and (3.1) bounds the remaining kernels and their local derivatives uniformly. For the exterior region put $x_i = e^{v_i}$, with lower endpoints $\ell_i = \log(1+\delta_i)$ and $\sum v_i = -\log q$. Translation and rescaling give a fixed simplex, with weight $\prod w_i^{-1/2}$ times a positive analytic function. Its total length is strictly positive on the stated domain, and this weight is integrable. Local differentiation under these fixed-domain integrals proves the claim. □

4 Quadratic transformations and first derivatives

Use the six row vectors

$$\begin{aligned} L_0 &= (1,0,-r), & L_1 &= (1,0,-1),\\ L_2 &= (-r,1,0), & L_3 &= (-1,1,0),\\ L_4 &= (0,-rT,1), & L_5 &= (0,-T,1). \end{aligned}$$

For a pairing $(i,j),(k,l),(m,n)$, form the matrix whose columns are the indicated cross products, and make the quadratic substitution

$$M = (L_i \times L_j,\ L_k \times L_l,\ L_m \times L_n), \qquad (x:s:z) = M(VW:WU:UV). \tag{4.1}$$

The determinant is nonzero in each neighborhood under consideration. Each transformed factor $L_\nu M(VW,WU,UV)^{\mathsf{t}}$ is a coordinate times a linear factor. The six coordinate factors multiply to $(UVW)^2$, and cancel against the absolute Jacobian in (2.4). The remaining factor from the transformation is $|\det M|$.

The normalization can be computed entirely from the original rows. For pair number j, entry j of $L_\nu M$ is zero; interchange the other two entries to obtain the new linear factor. For the three cases use coordinate orders $(U,V,W),(W,V,U),(U,W,V)$, respectively. The pairs omitting the second, third, and first coordinates have endpoints

$$e_1 = -v_3/v_1, \qquad e_2 = -v_1/v_2, \qquad e_3 = -v_2/v_3.$$

If their midpoints and positive half-widths are b_i, A_i, set $\delta_i = b_i/A_i$, $q = A_1A_2A_3$. If C is the product of the six coefficients used in these endpoint ratios, then

$$g^2 = \frac{(\det M)^2}{C}, \qquad g > 0. \tag{4.2}$$

The row multipliers are retained in C. Direct cancellation in (4.1) proves this formula, including the normalization factors.

Proposition 4.1. *The pairings*

$$\begin{array}{c|l} 1 & (0,5),(1,2),(3,4) \\ 2 & (0,2),(1,5),(3,4) \\ 3 & (0,3),(1,4),(2,5) \end{array}$$

give the data in table 1. In case 2 the first and third multiplicative variables are interchanged; in case 3 use $(u,v,z) \mapsto (-u,z,-v)$. *With* $h = q^2/64$, $\theta_h = h\,\mathrm{d}/\mathrm{d}h$, *and* $\partial_i = \partial/\partial\delta_i$, *the transformed weighted operators, after factoring out* g, *are*

$$\begin{aligned} \mathcal{D}_1 &= 2 + 12\theta_h - 30\partial_1 - 30\partial_2 + 30\partial_3, \\ \mathcal{D}_2 &= -\frac{16}{3} + \frac{45}{2}\partial_1 - \frac{56}{3}q\partial_q - 72\partial_2 - 72\partial_3, \\ \mathcal{D}_3 &= -470 - 3948\theta_h + 2730(\partial_1 + \partial_2 + \partial_3). \end{aligned} \tag{4.3}$$

Thus the corresponding original weighted sum is $g\mathcal{D}_i P$, *evaluated at the point in the table.*

Table 1: The actual periods and normalization factors.

Case	$\mathcal{F}_r/g$	$(\delta_1,\delta_2,\delta_3;q)$	g
1	P_-	$(0,0,0;1/2)$	$6\sqrt{6}/11$
2	P_0	$(-11/16,0,0;16/45)$	$80/63$
3	P_+	$(0,0,0;1/8)$	$54\sqrt{2}/95$

Proof. The endpoint rules above give the algebraic factors and their derivatives. For example, $\vartheta \log g = -5/22$ and $\vartheta \log h = 2/11$ in case 1; they are $-77/190, -94/95$ in case 3. In case 2, $\vartheta \log g = -109/378$. The product rule for $A\vartheta + B$ gives (4.3).

It remains to verify the actual integration regions in a neighborhood, so that these derivatives apply to the required real period. The old inequalities $L_0L_1 \le 0$, $L_2L_3 \le 0$ transform, with square denominators, into

$$\begin{array}{c|ll} 1 & y(1-qa_3^-xy)(x-a_1^-) \le 0, & y(x-a_1^+)(y-a_2^-) \ge 0, \\ 2 & x(y-a_2^-)(1-qa_3^+xy) \ge 0, & x(a_2^+-y)(x-a_1^+) \le 0, \\ 3 & x(a_2^+-y)(1-qa_3^-xy) \ge 0, & x(a_1^+-x)(y-a_2^-) \ge 0. \end{array} \tag{4.4}$$

The original nonzero coefficients retain their signs locally. In row 1, $y > 0$ gives no region, $a_2^- < y < 0$ gives H_-, and $y < a_2^-$ gives the $(+,-,-)$ exterior region. In row 2, $x < 0$ gives no region, $0 < x < a_1^+$ gives H_+, and $x > a_1^+$ gives J^+. In row 3 the same three ranges give, respectively, no region, H_+, and the $(+,-,-)$ exterior region. These statements follow by using the second inequality to locate the other variable, and then the first to truncate it. On central regions the remaining sign follows from (3.1). For example, in row 3 with $x > a_1^+$, one has $y < a_2^-$, and the remaining condition $1 - qa_3^-xy \ge 0$ is exactly $1/(qxy) < a_3^-$.

Birationality gives multiplicity one away from algebraic curves of measure zero. The absolute Jacobian and positive roots give the positive factor g. The local period identities may therefore be differentiated by theorem 3.1. The exact computations use the variable T symbolically; scripts are specified in Appendix B. □

5 Common elliptic period identities

Write

$$F(m) = {}_2F_1(1/2,1/2;1;m), \quad K(m) = \frac{\pi}{2}F(m), \quad R = \sqrt{1-q^2}, \quad \lambda = \frac{1-R}{2},$$

and define

$$B(h) = \sum_{n\geq 0} \binom{2n}{n}^3 h^n = F(\lambda)^2, \qquad B_c(h) = F(1-\lambda)^2, \qquad h = q^2/64. \tag{5.1}$$

The equality for B is Clausen's identity; equivalently it follows from $[\theta_h^3 - 64h(\theta_h+1/2)^3]B = 0$ and the initial value. Arcsine moments show that B is the complete central rectangle.

Lemma 5.1. *For $0 < q < 1$,*

$$P_\pm(q;0,0,0) = \frac{3B + B_c}{8}, \tag{5.2}$$

and the center derivatives, evaluated at zero centers, are

$$\nabla_\delta P_- = \frac{(-1,-1,1)}{2\pi R}, \qquad \nabla_\delta P_+ = \frac{(1,1,1)}{2\pi R}. \tag{5.3}$$

More generally, on the domain (3.1)*,*

$$\partial_1 J^+(q;0,0,\eta) = \partial_2 J^+(q;0,0,\eta) = -\frac{1}{2\pi\sqrt{(1-q\eta)^2 - q^2}}, \tag{5.4}$$

and, when $\delta = (\eta,0,0)$,

$$\partial_2 H_+ = \partial_3 H_+ = 0, \qquad \partial_2 P_0 = \partial_3 P_0 = -\frac{1}{2\pi\sqrt{(1-q\eta)^2 - q^2}}. \tag{5.5}$$

Proof. First substitute $s = xy$ in the exterior region and integrate in x:

$$J^+(q;0) = \frac{1}{\pi^2}\int_1^{1/q} \frac{K(1-s^{-2})}{s\sqrt{1-q^2s^2}}\,\mathrm{d}s = \frac{B_c - B}{8}. \tag{5.6}$$

To justify the last equality, put $d = 1-q^2$, then $u = 1/s$, $u^2 = 1 - dw$. The integral becomes

$$\frac{\sqrt{d}}{4\pi}\int_0^1 \frac{F(dw)}{\sqrt{1-w}\sqrt{1-dw}}\,\mathrm{d}w.$$

If $(1-v)^{-1/2}F(v) = \sum h_n v^n$, Gauss's equation gives

$$n^2 h_n - [2n(n-1) + 3/4]h_{n-1} + (n-1)^2 h_{n-2} = 0.$$

Beta integration gives the series $\sqrt{d}(2\pi)^{-1}\sum j_n d^n$, where $j_n = n!h_n/(3/2)_n$. Its differential equation is

$$d(1-d)^2 Y''' + \frac{3}{2}(1-d)(1-3d)Y'' + \frac{13d-9}{4}Y' + \frac{1}{8}Y = 0. \tag{5.7}$$

The symmetric square of Gauss's equation shows that $(B_c - B)/8$ satisfies the same equation. Its leading term at $d = 0$ is $\sqrt{d}/(2\pi)$, by Legendre's relation

$$F'(m)F(1-m) + F(m)F'(1-m) = \frac{1}{\pi m(1-m)}. \tag{5.8}$$

Indeed this gives $F(1/2)F'(1/2) = 2/\pi$. The recurrence for a solution of the form $\sqrt{d}$ times an analytic series has nonzero leading coefficient $n(n^2-1/4)$ for $n \geq 1$. The leading term therefore determines that solution uniquely. Real analytic continuation proves (5.6) for all

$0 < q < 1$. The central half-rectangle is $B/2$, proving (5.2). For completeness, (5.8) follows from the Wronskian of Gauss's equation and $F(1-m) = \pi^{-1}\log(16/m) + O(m\log m)$.

To compute center derivatives while retaining the moving boundary, put

$$f_\delta(v) = \frac{\mathbf{1}_{\{v>\log(1+\delta)\}}}{\sqrt{(e^v-\delta)^2-1}}, \qquad L = -\log q.$$

Then

$$J^+(q;\delta) = \frac{(f_{\delta_1} * f_{\delta_2} * f_{\delta_3})(L)}{\pi^2 q}.$$

For Laplace parameter $s > 0$, a single factor transforms to

$$\mathcal{L}_\delta(s) = \int_0^\infty (\delta + \cosh t)^{-s-1}\,\mathrm{d}t, \qquad \partial_\delta \mathcal{L}_\delta(s) = -(s+1)\mathcal{L}_\delta(s+1),$$

and a beta integral gives $\mathcal{L}_0(s)\mathcal{L}_0(s+1) = \pi/[2(s+1)]$. Thus, at $(0,0,\eta)$, the first or second center derivative of the convolution is $-\pi f_\eta(L)/2$. Laplace uniqueness applies: after shifting its support endpoint the convolution is a square root times an analytic function, its derivative has at worst an inverse square-root singularity, and it decays exponentially at infinity. There is no endpoint point mass. This proves (5.4).

At zero centers, $\nabla H_- = (0, -1/(\pi R), 0)$ and $\nabla H_+ = (1/(\pi R), 0, 0)$. In the truncated coordinate the angular boundary contributes $\pm 1/\pi$, and the interior derivative contributes $\pm(R^{-1}-1)/\pi$. The other derivatives vanish by oddness. The exterior $(+,-,-)$ derivative is $(-1,1,1)/(2\pi R)$, by (5.4) and the sign substitution. This proves (5.3). The same oddness in an untruncated zero-center coordinate gives (5.5). □

6 Evaluation of the three series

We use the following classical evaluations:

$$(6\theta_h + 1)B(1/256) = 4/\pi, \qquad (42\theta_h + 5)B(1/4096) = 16/\pi, \tag{6.1}$$

and

$$\begin{aligned} F(1-\lambda_3)/F(\lambda_3) &= \sqrt{3}, \quad \lambda_3 = (2-\sqrt{3})/4, \\ F(1-\lambda_7)/F(\lambda_7) &= \sqrt{7}, \quad \lambda_7 = (8-3\sqrt{7})/16. \end{aligned} \tag{6.2}$$

The first two identities have WZ proofs in [1, 2]. The degree-three singular value is given in [3, Lemmas 2.10–2.11]; its modulus must be squared to obtain the present parameter. The degree-seven value and the relevant elliptic relations are given in [4, Section 9].

At a point where $F(1-\lambda)/F(\lambda) = \sqrt{N}$, (5.8) gives

$$\theta_h B_c = N\theta_h B - \frac{2\sqrt{N}}{\pi R}. \tag{6.3}$$

This is obtained from the two independent function derivatives and the Wronskian relation; it does not differentiate a special-value ratio valid only at one point.

Proof of theorem 1.1. For case 1 set $q = 1/2$, $R = \sqrt{3}/2$, $h = 1/256$. Equations (5.2), (6.1)–(6.3) give

$$(2 + 12\theta_h)P_- = \frac{1}{8}\{12B + 72\theta_h B - 48/\pi\} = 0.$$

The center-derivative part of $\mathcal{D}_1 P_-$ is $90/(\pi\sqrt{3})$, by (5.3). Multiplication by $g = 6\sqrt{6}/11$ proves (1.1).

For case 3 set $q = 1/8$, $R = 3\sqrt{7}/8$, $h = 1/4096$. The same equations give

$$\begin{aligned}(-470 - 3948\theta_h)P_+ &= \frac{1}{8}\{-4700B - 39480\theta_h B + 7896\sqrt{7}/(\pi R)\} \\ &= 752/\pi.\end{aligned}$$

The remaining center-derivative contribution is $8190/(2\pi R)$. Multiplying both contributions by $54\sqrt{2}/95$ gives $432 \cdot 94\sqrt{2}/(95\pi)$ and $432 \cdot 195\sqrt{14}/(95\pi)$, respectively. This proves (1.3).

For case 2 put $P(q,\eta) = P_0(q;\eta,0,0)$. Appendix A proves the exact evaluation

$$\left[-\frac{16}{3} + \frac{45}{2}\partial_\eta - \frac{56}{3}q\partial_q\right] P(16/45, -11/16) = \frac{6\sqrt{5}}{\pi}. \tag{6.4}$$

At this point $\sqrt{(1-q\eta)^2 - q^2} = 8\sqrt{5}/15$. The two normal derivatives in $\mathcal{D}_2$, after multiplication by $80/63$, give $720\sqrt{5}/(21\pi)$, by (5.5). The tangential part (6.4), multiplied by $80/63$, gives $160\sqrt{5}/(21\pi)$. Their sum proves (1.2). Absolute convergence was established in section 2. □

A The second period evaluation

The symbols P, q, η, F, K retain their definitions from the main text.

In this appendix F is its ordinary analytic branch on the complex plane cut along $[1,\infty)$, and $F_c(z) = F(1-z)$, with values continued on the explicitly indicated paths. $K(z) = \pi F(z)/2$ retains the elliptic parameter convention.

A.1 The actual positive-partial-rectangle plus exterior period

For $q > 0$, $|\eta| < 1$ and $q(1+|\eta|) < 1$ set

$a = q(1+\eta)$, $\quad b = q(1-\eta)$, $\quad B = q\eta$, $\quad D = B^2 - q^2 = -ab$.

Define

$$P(q,\eta) = \frac{2}{\pi^2}\int_0^a \frac{K(U^2)\,dU}{\sqrt{(a-U)(U+b)}} + \frac{1}{\pi^2}\int_a^1 \frac{K(1-U^2)\,dU}{\sqrt{(U-a)(U+b)}}. \tag{A.1}$$

Here $U = qx$, where x is the normalized first variable of Section 3; thus its central interval $[\eta-1, \eta+1]$ becomes $[-b, a]$. Both integrals converge and have positive real square roots. This is exactly the chamber family Section 4 identifies for case 2 after permuting the third moment variable: a positive part $0 < U < a$ of the first arcsine interval $[-b, a]$, the whole second interval $[-1, 1]$, plus the positive exterior chamber $a < U < 1$, $1 < V < 1/U$. The second inner integration is $K(1-U^2)$; the first is $2K(U^2)$.

A.2 Algebraic elliptic parameters

Set

$$\Delta = \sqrt{(1+D)^2 - 4B^2}, \quad w = \sqrt{(1+D+\Delta)/2}, \quad \rho = \sqrt{D} = iq\sqrt{1-\eta^2},$$

with Δ, w positive on the stated real domain. Indeed $\Delta^2 = (1-a^2)(1-b^2) > 0$ and $w > |B|$. Put

$$\begin{gathered}X = \frac{w(w+\rho)}{w+B}, \quad Y = \frac{w(w-\rho)}{w+B}, \\ \lambda_1 = \frac{2 - X - X^{-1}}{4}, \quad \lambda_2 = \frac{2 - Y - Y^{-1}}{4}, \quad G = \frac{1+XY}{2\sqrt{XY}}.\end{gathered} \tag{A.2}$$

Here $XY > 0$, and $\sqrt{XY} > 0$. X lies in the upper right quadrant and $Y = \overline{X}$; the λ_i are conjugate. At $\eta = 0$,

$$X = R + iq, \quad Y = R - iq, \quad XY = 1, \quad G = 1, \quad \lambda_1 = \lambda_2 = \lambda = \frac{1-R}{2}, \quad R = \sqrt{1-q^2}.$$

Direct substitution verifies

$$B = \frac{(X+Y)(1-XY)}{(1+XY)^2}, \qquad D = \frac{(X-Y)^2}{(1+XY)^2}. \tag{A.3}$$

A.3 Exact product formula for the actual period

Proposition A.1 (Product formula)**.**

$$\boxed{\begin{aligned} P = \frac{G}{8}\big[&3F(\lambda_1)F(\lambda_2) + F_c(\lambda_1)F_c(\lambda_2) \\ &- i\{F(\lambda_1)F_c(\lambda_2) - F_c(\lambda_1)F(\lambda_2)\}\big]. \end{aligned}} \tag{A.4}$$

Proof. The anti-symmetric mixed term is essential for first derivatives. Replacing P by half a fundamental series would lose it.

A.3.1 The common Euler–Poisson–Darboux equation

At fixed B,q the kernel $k(U) = [(U-B)^2 - q^2]^{-1/2}$ satisfies pointwise

$$(\partial_B^2 - \partial_q^2 - q^{-1}\partial_q)k = 0. \tag{A.5}$$

For P, use fixed-contour integrals $I_1 = \int_0^1 K(U^2)k(U)\,dU$ and $I_2 = \int_0^1 K(1-U^2)k(U)\,dU$, with the root chosen as $+i$ times the positive root on $0 < U < a$ and positive on $a < U < 1$. Then

$P = (-2\,\mathrm{Im}\, I_1 + \mathrm{Re}\, I_2)/\pi^2$

on real parameters. More precisely real and imaginary parts are constant linear combinations of the holomorphic upper/lower contour branches. Off their cut, differentiate the fixed contours using (A.5); their endpoint logarithms are integrable. Continue to the indicated local boundary branches. Thus P satisfies (A.5) as a real analytic function. In (q, η) coordinates this is

$$(1-\eta^2)P_{\eta\eta} + 2q\eta P_{q\eta} - q^2 P_{qq} - qP_q - \eta P_\eta = 0. \tag{A.6}$$

There is no moving-endpoint differentiation assumption in this argument: all contour endpoints are 0 and 1 before continuation.

Every product on the right of (A.4), including its gauge, satisfies the same equation. An explicit exact verification is `n2_epd_certificate.py` with output `n2_epd_certificate.json`. In coordinates (B, D), equation (A.5) is

$$\mathcal{L} = \partial_B^2 + 4B\partial_B\partial_D + 4D\partial_D^2 + 6\partial_D.$$

For any solutions $h(X), j(Y)$ of

$$X(1-X^2)h'' + (1-3X^2)h' - Xh = 0 \tag{A.7}$$

and its Y analogue, the certificate proves identically

$$\mathcal{L}\left(\frac{1+XY}{2}h(X)j(Y)\right)=0.$$

It transforms the derivatives using the inverse Jacobian of (A.3), substitutes (A.7), and obtains exactly zero for all four remaining coefficients $h'j', h'j, hj', hj$. The transformed mixed second-derivative coefficient is already zero. This is a finite symbolic certificate of a rational identity, not finite testing of parameter values.

The elementary quadratic transformation

$F(1-Z^2)=Z^{-1/2}F((2-Z-Z^{-1})/4)$

shows that both $Z^{-1/2}F(\lambda(Z))$ and $Z^{-1/2}F_c(\lambda(Z))$ solve (A.7); this also follows directly from the Gauss equation. Hence all four product branches in (A.4) satisfy (A.6).

A.3.2 Cauchy data on $\eta=0$

By the complete outer-chamber evaluation proved in Section 5, equations (5.2)–(5.7),

$$P(q,0)=\frac{3F(\lambda)^2+F_c(\lambda)^2}{8}. \tag{A.8}$$

Moreover

$$P_\eta(q,0)=\frac{1}{2\pi R}. \tag{A.9}$$

Here is a genuine moving-domain calculation of (A.9). The positive partial rectangle is

$$H_+=\frac{1}{\pi^2}\int_0^{\arccos(-\eta)}\int_0^{\pi}\frac{d\phi\,d\theta}{\sqrt{1-q^2(\eta+\cos\theta)^2\cos^2\phi}}.$$

At $\eta=0$ its moving upper boundary contributes $1/\pi$. Differentiating the uniformly bounded interior kernel, then putting $z=\sin\theta$, gives the additional contribution

$$\frac{1}{\pi^2}\int_0^{\pi}\frac{q^2\cos^2\phi}{1-q^2\cos^2\phi}\,d\phi=\frac{R^{-1}-1}{\pi}.$$

Thus $(H_+)_\eta=1/(\pi R)$. The symmetric three-positive exterior period has center derivative $-1/(2\pi R)$, giving (A.9). A complete ordinary-integral proof of that exterior derivative is Section 5, equation (5.4): after logarithmic coordinates its Laplace transform is the product of $\mathcal{L}_\delta(p)=\int_0^\infty(\delta+\cosh t)^{-p-1}\,dt$, and $\mathcal{L}_0(p)\mathcal{L}_0(p+1)=\pi/[2(p+1)]$, so the derivative is $-1/(2\pi R)$. The support and endpoint estimates required for differentiating and inverting are given in Section 5.

For the right of (A.4), direct differentiation of (A.2) at $\eta=0$ gives

$$G_\eta=0,\quad (\lambda_1)_\eta=\frac{iq^2}{2R},\quad (\lambda_2)_\eta=-\frac{iq^2}{2R}.$$

The symmetric terms have zero η derivative; the anti-symmetric term has derivative $4i/(\pi R)$ by the elliptic Legendre relation

$$F'(z)F(1-z)+F(z)F'(1-z)=\frac{1}{\pi z(1-z)}, \tag{A.10}$$

and $\lambda(1-\lambda)=q^2/4$. Thus it gives exactly (A.9), and its value gives (A.8).

Equation (A.6) is noncharacteristic in η at $\eta=0$. Analytic uniqueness (Cauchy–Kowalevskaya, or recursively solving for all η Taylor coefficients in (A.6)) proves (A.4) near that line. Both sides are real analytic on the connected domain $q>0,\ |\eta|<1,\ q(1+|\eta|)<1,$

with the stated branches, so the identity continues throughout that domain, including the case 2 point.

□

A.4 The required exact tangential evaluation

Lemma A.2 (Tangential evaluation)**.** *At*

$$q_0 = 16/45, \quad \eta_0 = -11/16, \quad \mathscr{V} = \frac{45}{2}\partial_\eta - \frac{56}{3}q\partial_q,$$

we prove

$$\boxed{\left(-\frac{16}{3} + \mathscr{V}\right) P = \frac{6\sqrt{5}}{\pi}.} \tag{A.11}$$

Proof.

All remaining algebra is certified exactly by `case2_exact_tangent.py` and `case2_exact_tangent.json` in $\mathbb{Q}(i, \sqrt{3}, \sqrt{5})$; the short identities below explain what is certified.

Define

$$x = \frac{2\sqrt{5} - 3 + i(\sqrt{15} - 2\sqrt{3})}{2}, \quad y = \overline{x}, \quad xy = 14 - 6\sqrt{5}, \quad x^2 + y^2 = 1.$$

At the target, $X = 1/y,\ Y = 1/x$ in (A.2), so, writing $l = \lambda_1,\ n = \lambda_2$,

$$l = \frac{47 - 21\sqrt{5} + i(7\sqrt{15} - 17\sqrt{3})}{64}, \quad n = \overline{l} = \frac{l}{l-1}. \tag{A.12}$$

Let $A = F(l),\ C = F(1-l),\ A' = F'(l),\ C' = F'(1-l)$, and put $t = (\sqrt{15} + i)/2$. We first establish the value and derivative relations

$$\boxed{C = tA, \qquad C' = -C_1 A' - C_0 A,} \tag{A.13}$$

where

$$C_1 = \frac{-\sqrt{15} + i}{2}, \qquad C_0 = \frac{3\sqrt{3} - i(32 + 17\sqrt{5})}{8}. \tag{A.14}$$

These are proved by differentiating an actual local modular correspondence, not an isolated special-value ratio.

A.4.1 Quadratic and Pfaff connection formulas with phases

For Z in the upper right quadrant near our paths, with $\Lambda(Z) = (2 - Z - Z^{-1})/4$,

$$F(1 - Z^2) = Z^{-1/2} F(\Lambda(Z)), \quad F(1 - \Lambda(Z)) = \sqrt{Z}\,[2F(Z^2) - iF(1 - Z^2)]. \tag{A.15}$$

For the lower right quadrant the second sign is $+i$. These are elementary Gauss quadratic/Pfaff connection formulas. They may be verified without a table: substitute into the Gauss equation, use the analytic-at-$Z = 1$ solution for the first, and match the logarithm and constant of $F(1 - \Lambda)$ at $Z = 1$ from above/below for the second. The standard local expansion $F(1 - z) = \pi^{-1} \log(16/z) + O(z \log z)$ fixes exactly the displayed sign. All paths here remain in the indicated quadrants and avoid the hypergeometric cuts.

The corresponding Pfaff identities at $n = l/(l-1),\ \operatorname{Im} l < 0$ are

$$F(n) = sA, \qquad F(1-n) = s(C - iA), \quad s = \sqrt{1-l}. \tag{A.16}$$

The first is Pfaff's transformation; the second is its complementary connection formula, fixed by the same logarithmic expansion on the lower half-plane. Differentiating the first identity as a functional identity in l gives

$$F'(n) = -s^5 A' + \frac{s^3}{2} A. \tag{A.17}$$

No claim about the motion of n in the original q,η direction is needed for (A.17): it is the value of F' at this argument, obtained from the transformation itself.

A.4.2 Value multiplier, and a legitimate derivative correspondence

Take $f = F(x^2),\ g = F(y^2)$. Since $x^2 + y^2 = 1$, (A.15) gives

$$A = \sqrt{y}\, f, \quad C = \sqrt{y}(2g + if), \quad F(n) = \sqrt{x}\, g, \quad F(1-n) = \sqrt{x}(2f - ig). \tag{A.18}$$

Comparing the ratios in (A.16) and (A.18) and writing $r = g/f$ yields $2/r - i = 2r$, hence $2r^2 + ir - 2 = 0$. The Euler integral for $F(x^2)$, where $\mathrm{Re}(x^2) = 1/2$ and $\mathrm{Im}(x^2) > 0$, lies strictly in the sector $0 < \arg f < \pi/4$; $g = \overline{f}$, so $\mathrm{Re}(g/f) > 0$. Thus $r = (\sqrt{15} - i)/4$, giving $C/A = 2r + i = t$. This proves the first relation in (A.13).

For its derivative, vary x locally and put $y = \sqrt{1 - x^2}$ through the target. Set $l = \Lambda(y),\ n = \Lambda(x),\ k = \sqrt{x/y}$, with the continued roots. Relations (A.18) hold along this algebraic curve, yielding

$$F(n) = \frac{k}{2}(C - iA). \tag{A.19}$$

On this curve

$$h := \frac{dn}{dl} = -\frac{y^5}{x^5}, \qquad j := \frac{d\log k}{dl} = -\frac{2y}{x^4}. \tag{A.20}$$

Differentiate (A.19), use (A.17) at the target, and use $s/k = (t - i)/2$ there. The result is the second relation in (A.13), with

$$C_1 = i - h(1-l)^2(t-i), \quad C_0 = (t-i)\big(h(1-l)/2 - j\big).$$

Exact substitution of x,y reduces these to (A.14). This is the necessary multiplier-derivative step; differentiating only $C = tA$ at one isolated point would be invalid and is not used.

A.4.3 Final cancellation

Substitute (A.13), (A.16), and (A.17) into the derivative of the product expression (A.4). The exact algebraic certificate gives the particularly short result

$$\left(-\frac{16}{3} + \mathscr{V}\right) P = 6\sqrt{5}\, l(1-l)\left(\sqrt{15}\, AA' - C_0 A^2\right). \tag{A.21}$$

For checking this reduction, the certificate uses $G = (15 - 3\sqrt{5})/8$,

$$\mathscr{V} \log G = 28/3 - 6\sqrt{5},$$

$$\Psi l = \frac{-55 + 41\sqrt{5} + i(113\sqrt{15} - 235\sqrt{3})}{16}, \quad \Psi n = \overline{\Psi l}.$$

These follow by differentiating the rational map (A.3), with B replaced by -B after taking $x = 1/Y,\ y = 1/X$; the retained script calculates them exactly, not from decimal fitting. It formally expands the derivative in A, C, A', C', substitutes (A.13), and checks all coefficients.

Finally (A.10) and (A.13) give

$$AA't + A(-C_1A' - C_0A) \ = \ (t - C_1)AA' - C_0A^2 = \frac{1}{\pi l(1-l)}.$$

Since $t - C_1 = \sqrt{15}$, (A.21) is exactly (A.11).

□

B Exact algebraic certificates

The ancillary directory contains the exact scripts used for the finite algebraic parts of the proof. They use integer, rational, or $\mathbb{Q}(\mathrm{i}, \sqrt{3}, \sqrt{5})$ arithmetic. They do not establish convergence, identify a real integration region, or replace analytic continuation; those steps were proved in the text.

1. `cremona_gauge.py` and `case2_gauge.py` construct the six original lines, normalized endpoints, positive half-widths, gauge factors, and the full first derivative data.
2. `hostile_special_check.py` and `hostile_case2_check.py` check the branch factors, Jacobian, region inequalities, and the inverse-Jacobian calculation.
3. `n2_epd_certificate.py` and `brafman_epd_certificate.py` verify the four coefficient identities in the Euler–Poisson–Darboux reduction.
4. `case2_exact_tangent.py` verifies the target algebraic numbers, transformation derivatives, and the final cancellation.

The reduction of the differential equation is an identity in its free parameters after second derivatives are eliminated using Gauss's equation. The tangential evaluation is checked by comparing coefficients of the formal period products. These are exact symbolic checks.

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